+ve

1000 1002 1004

M T W

1000
1004
1002

Attention

food is delicious but price high] sentiment ?

Problem Statement

ASPECT BASED SENTIMENT ANALYSIS

```
INPUT = {
'text': 'food is delicious but price is high',
'aspect':'food',
}
```

input statement
input aspect

```
OUTPUT = {
'aspect_Sentiment' : 'Positive'
}
```

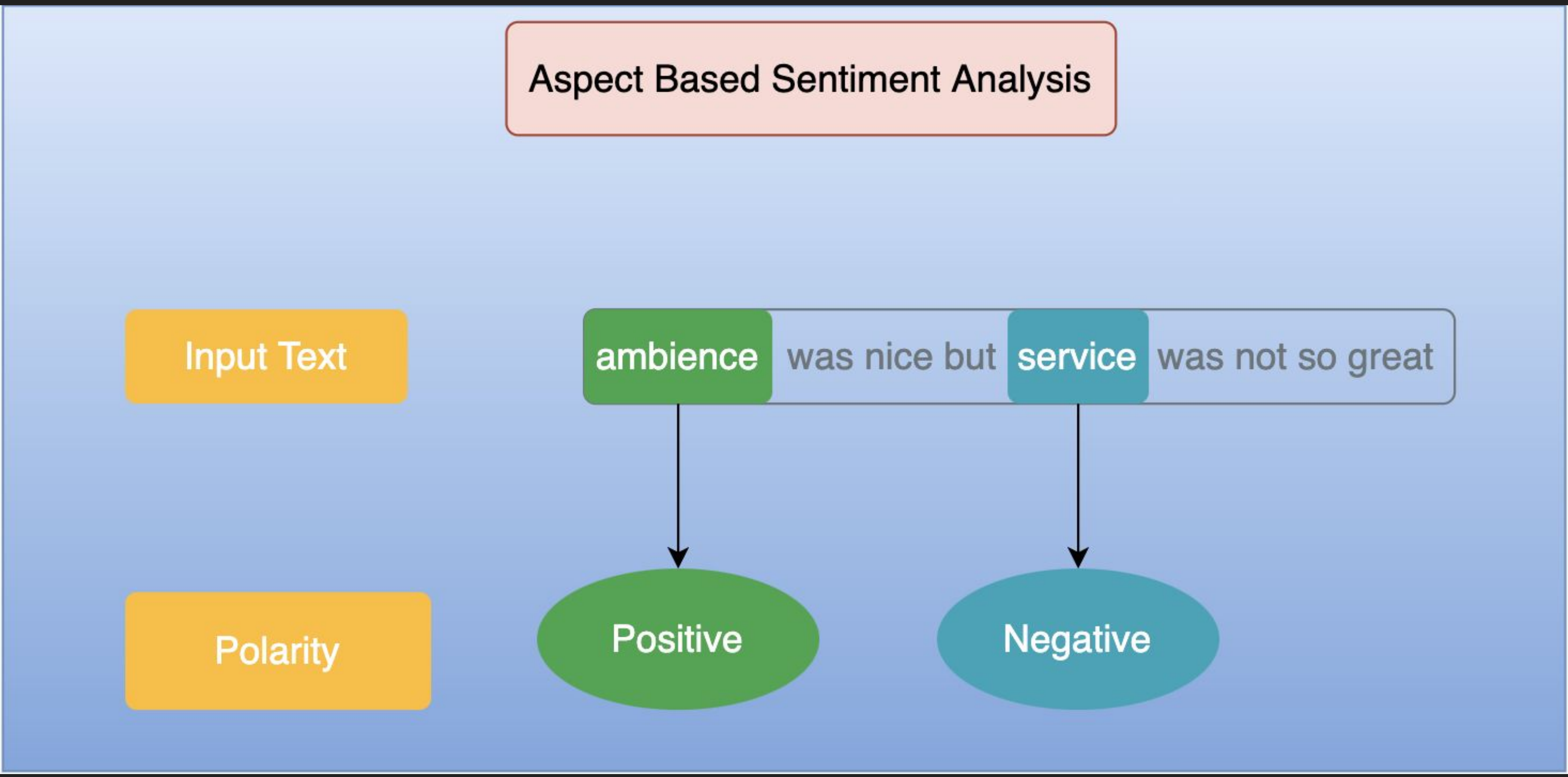
lunch

+ve

food is delicious price is high

-ve

cost


Aspect Based Sentiment Analysis
Input Text
ambience was nice but service was not so great
Polarity
Positive
Negative

How to approach the problem

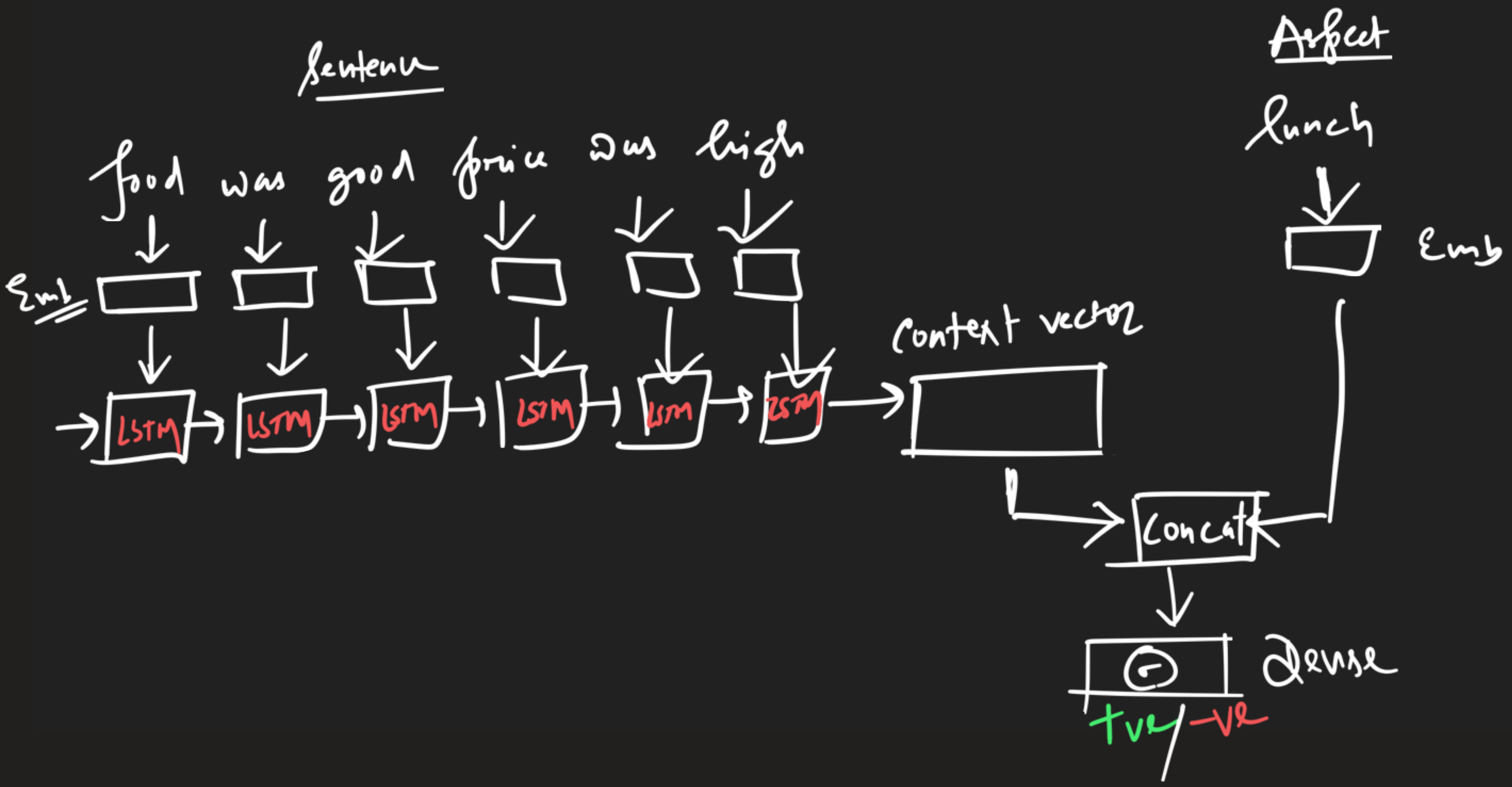
1] LSTM

2] $\underbrace{\text{LSTM + Attention}}$

no need of LSTM ⟹ Transformers

* LSTM treats entire sentence equally (confused with multiple sentiments)

LSTM based



The robber is at the Bank
The fish is at the river bank.

LSTM+Attention

$[0,1] \longleftarrow$ Range $[-\infty, \infty]$

food $\longleftarrow$ Query (Q)

$[\quad]_{1\times d}$

$P_1 = 0.9$ softmax(W1) P_2 softm(W2) P_3 softm W3 P_4 softm W4 P_5 softm W5 P_6 softm W6

Key(K) $\longrightarrow$ $[\quad]_{1\times d}$ lunch $[\quad]_{1\times d}$ was $[\quad]_{1\times d}$ delicious $[\quad]_{1\times d}$ price $[\quad]_{1\times d}$ was $[\quad]_{1\times d}$ high

$Q \cdot K^T$

$[\,]_{1\times d} \cdot [\,]^T_{6\times d}$

$[\,]_{1\times d} \cdot [\,]_{d\times 6}$

$[\quad]_{1\times 6}$

$softmax(Q \cdot K^T) = [\quad]_{1\times 6}$

Values $6\times d$

$\underset{0.5}{P_1} * [\,]_{1\times d}$ lunch $+ \underset{0.01}{P_2} * [\,]_{1\times d}$ was $+ \underset{0.2}{P_3} * [\,]_{1\times d}$ delicious $+ \underset{0.01}{P_4} * [\,]_{1\times d}$ price $+ \underset{0.0\cdot}{P_5} * [\,]_{1\times d}$ was $+ \underset{0.1}{P_6} * [\,]_{1\times d}$ high $\longrightarrow [\,]_{1\times d}$

Context vector

$V = K \}_{6\times d}$

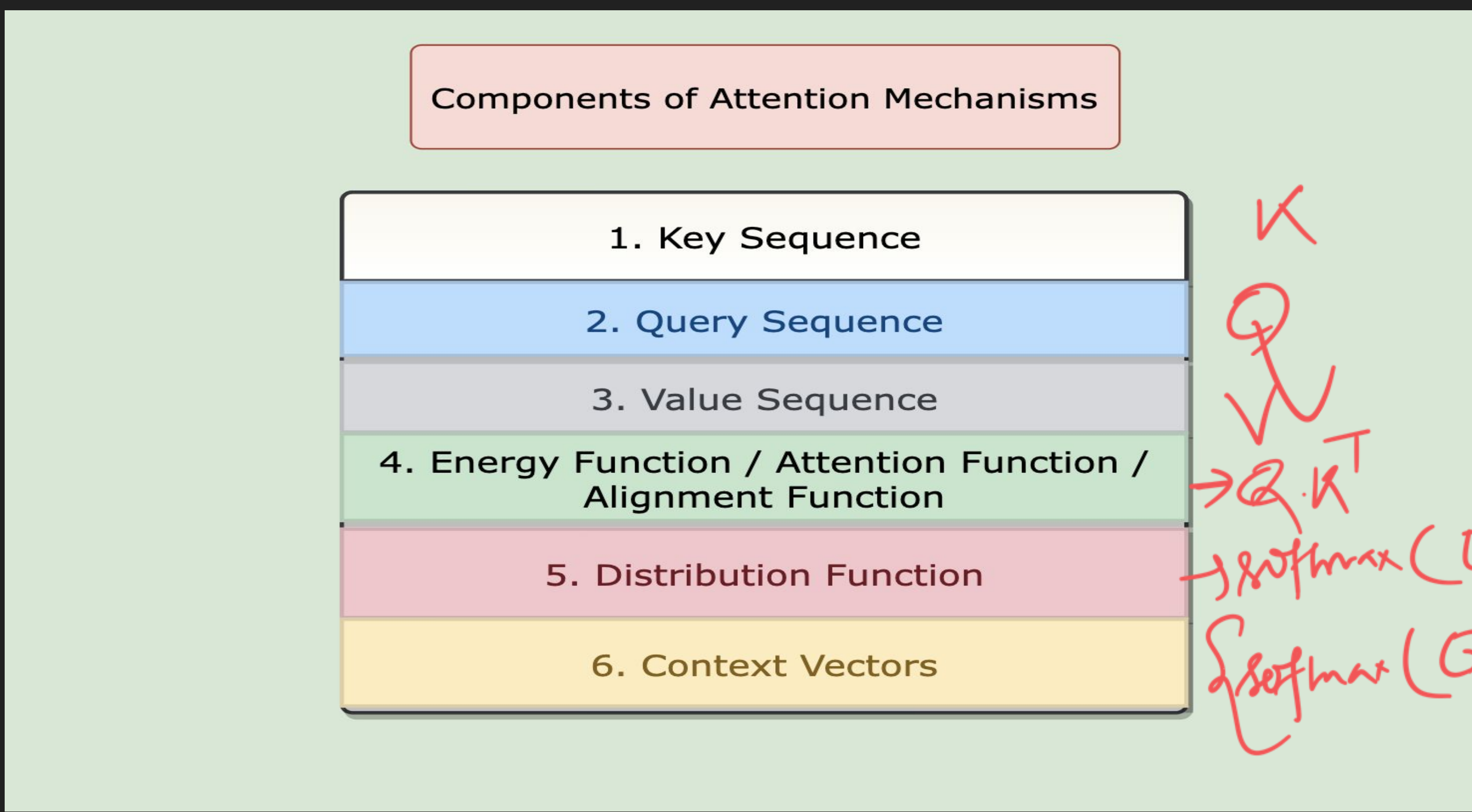
$softmax(Q K^T) * V$

input sentence

$$\text{softmax} \left(Q \cdot K^T \right) * V$$

aspect

Mathematical Representation of Attention



K

Q

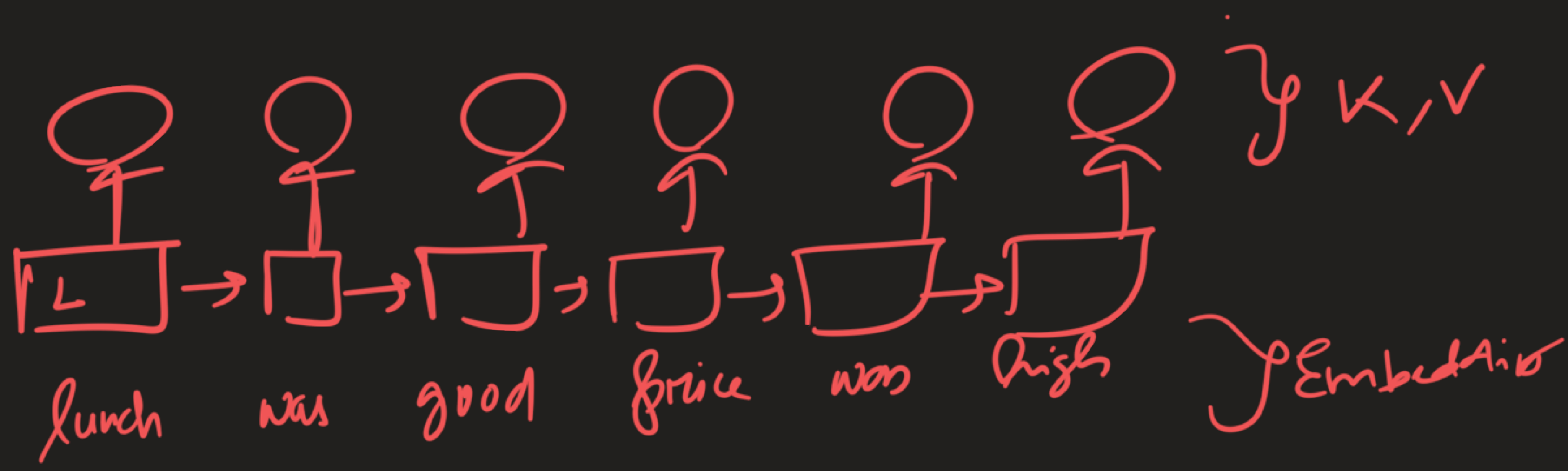
V

$\rightarrow Q \cdot K^T$

$\rightarrow \text{softmax}(Q \cdot K^T)$

$\{\text{softmax}(Q \cdot K^T) * V\}$

$\text{softmax}(Q \cdot K^T)$



$$attention\_score = Attention\_function(Q,\ K)$$
$$attention\_distribution = softmax(attention\_score)$$
$$Context\_vectors = attention\_distribution \times V$$

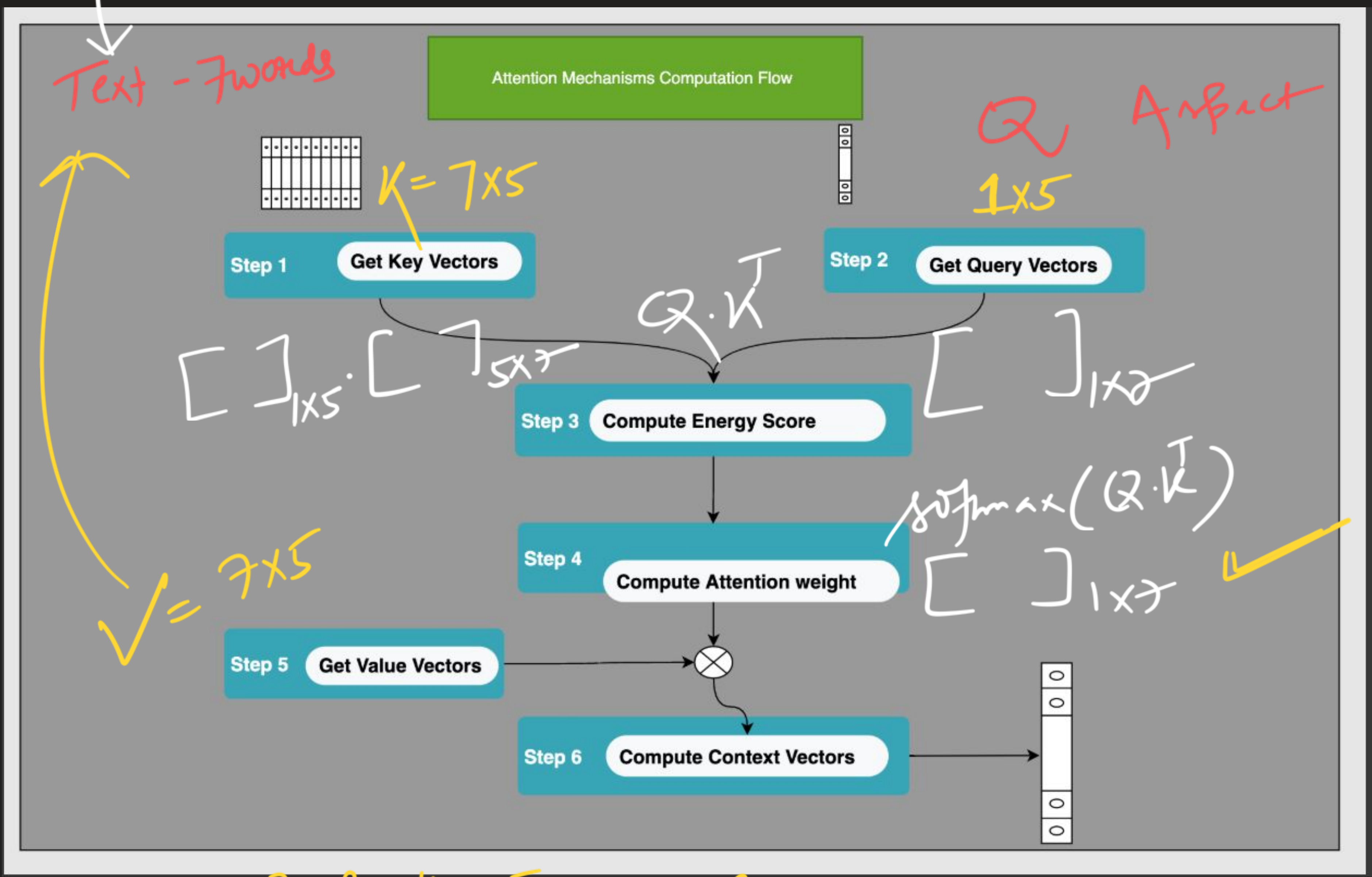
where,

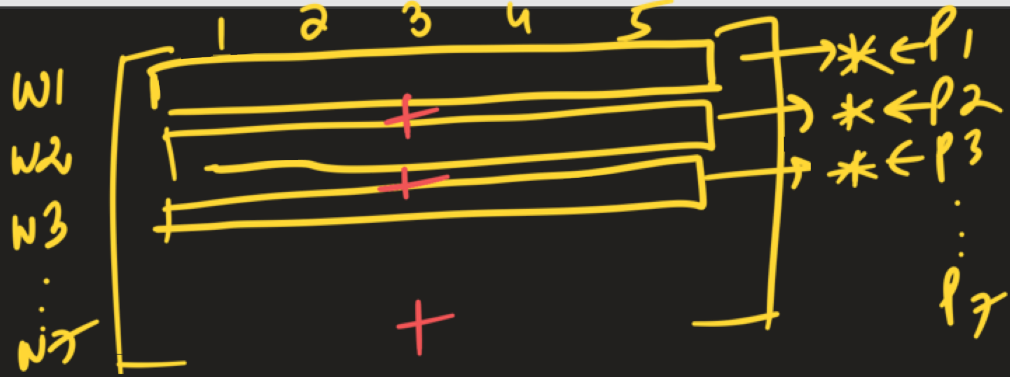
$$K = Key\ vector$$
$$Q = Query\ vector$$
$$V = Value\ vector$$
$$attention\_score = energy\_score$$
$$attention\_distribution = attention\_weights$$

For Our Task,
K = LSTM(Input Sequence)
Q = Embedding(aspect_term) → $Q = E_{aspect}$
V = K

7 words – each word $1 \times d$

7×5





size of context vector → 1×5

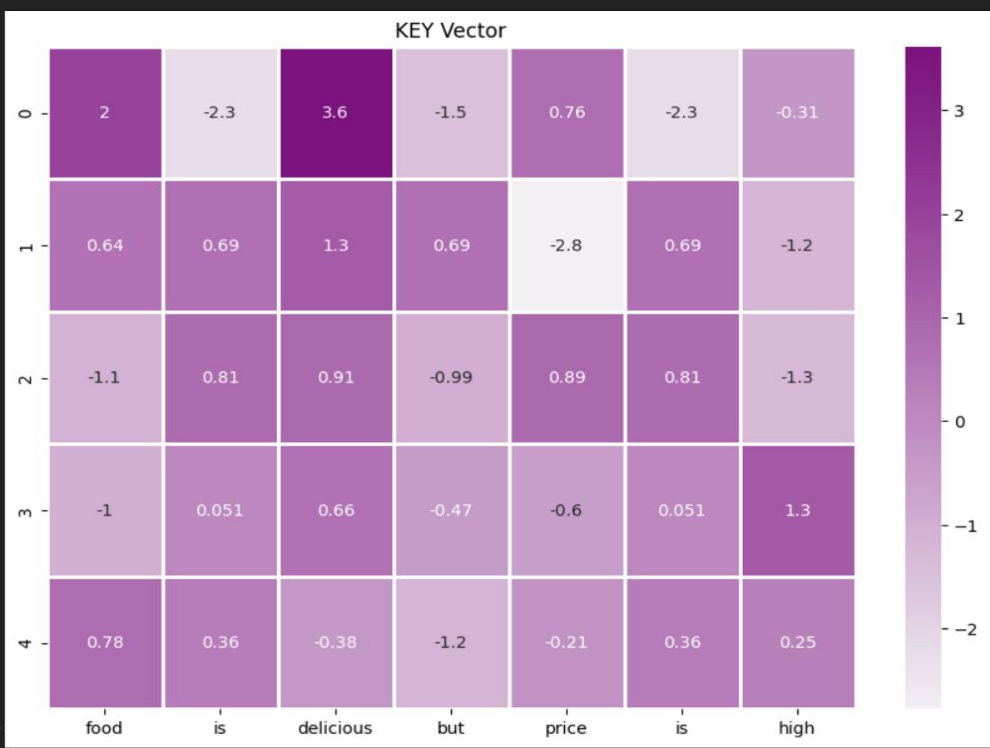
Example

Review text (sentence)

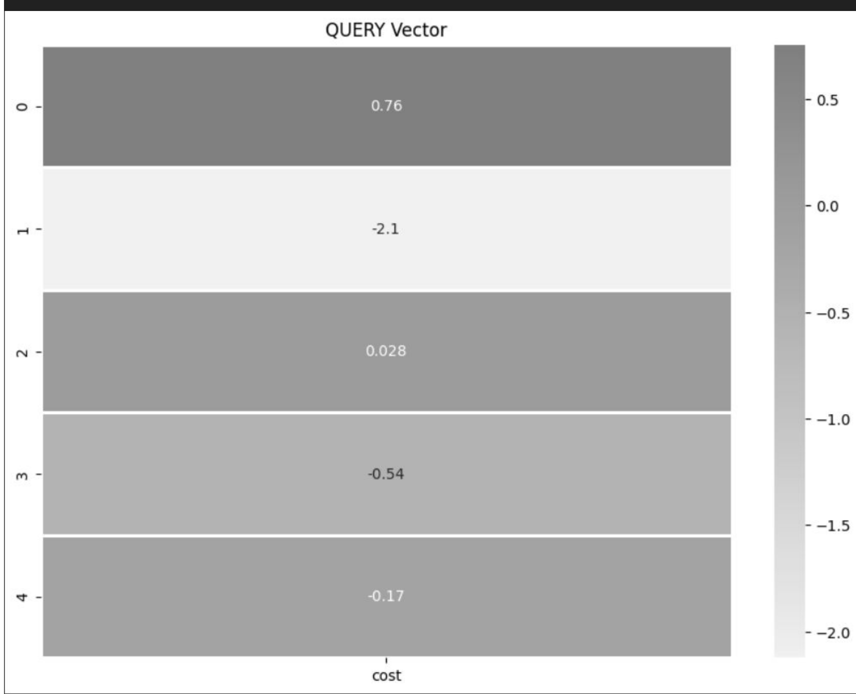
K = V

7x5

Q 1x5



sentence→



Energy

Attn distributi

$$\text{softmax}(Q \cdot K^T)$$

$$[\ ]_{1\times7} \quad - [\ ]_{K\times7}$$

$$[Q_{1\times5} \cdot K^T_{5\times7}]_{1\times7}$$

ENERGY_SCORE

```
w1 array([[ 0.51131411],
w2        [-3.23661494],
w3        [-0.24969932],
w4        [-2.17613397],
w5        [ 6.85134696],
w6        [-3.23661494],
w7        [ 1.53640209]])
```

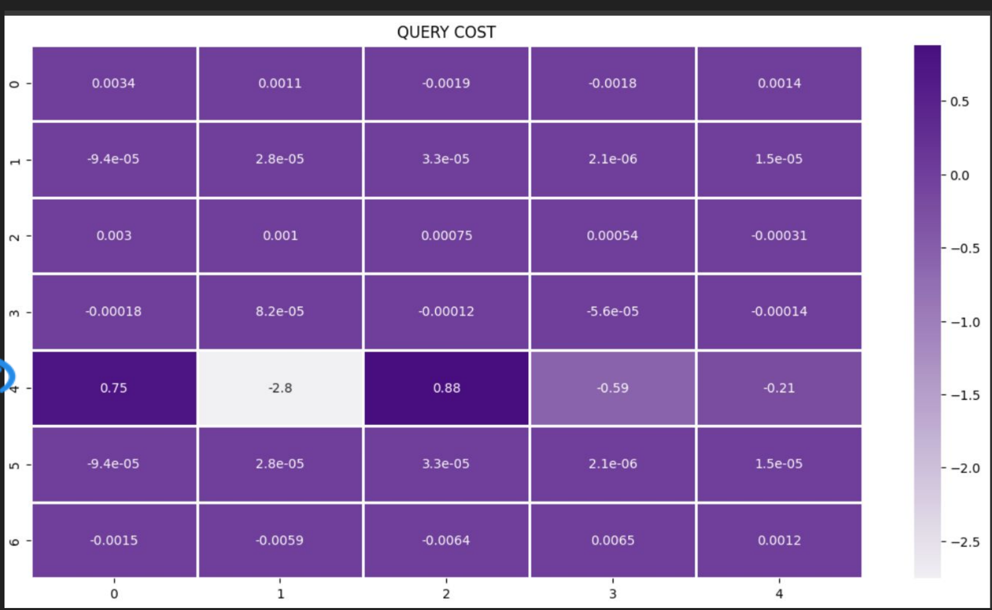
price

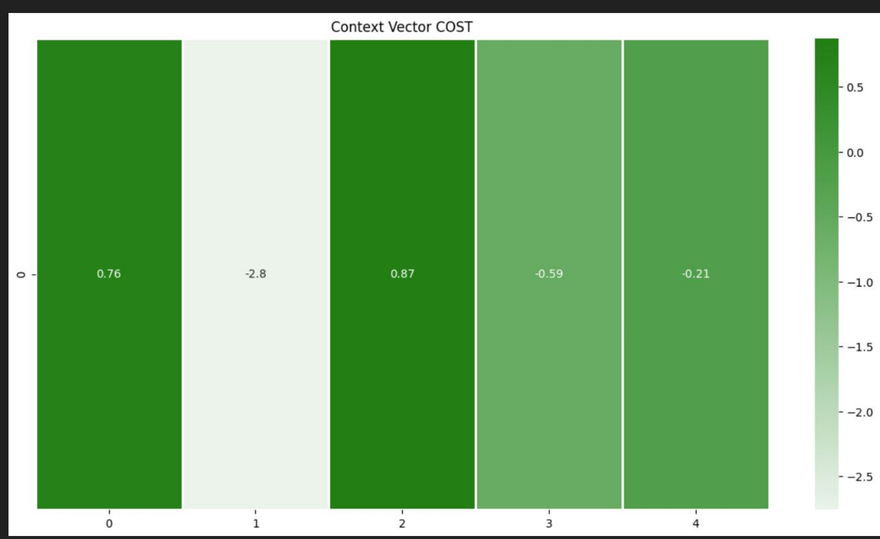
$W_0[\ ]_{1\times5} * P_0 + W_1[\ ]_{1\times5} * P_1 + \dots$

Context vector = 1x5

[

Cost





19:15 ✓

$\text{softmax}(Q \cdot K^T) \cdot V$

$P_1 \ * \ [\quad]_{1\times5}$ +

$P_2 \ * \ [\quad]_{1\times5}$ +

$P_3 \ * \ [\quad]_{1\times5}$ +

0.99 $P_4 \ * \ [\quad]_{1\times5}$ +

$P_5 \ * \ [\quad]_{1\times5}$ +

$P_6 \ * \ [\quad]_{1\times5}$ +

$P_7 \ * \ [\quad]_{1\times5}$ +

Context vector

— $[\quad]_{1\times5}$

extremely large/extremely

$$\text{softmax}\left(\frac{\overbrace{Q \cdot K^T}}{\sqrt{d}}\right)$$

size of embedding vector

$Q \cdot K$ vector

Scaled Dot Prod

$\text{softmax}([0.2 \quad 0.3])$

$\quad 0.4 \quad 0.6$

$\text{softm}([20 \quad . \quad 30])$

$0.01 \quad 0.99$

$-20 \qquad -0.2$

$Q \cdot K^T \quad -ve$

$Q \cdot K^T$

learnable

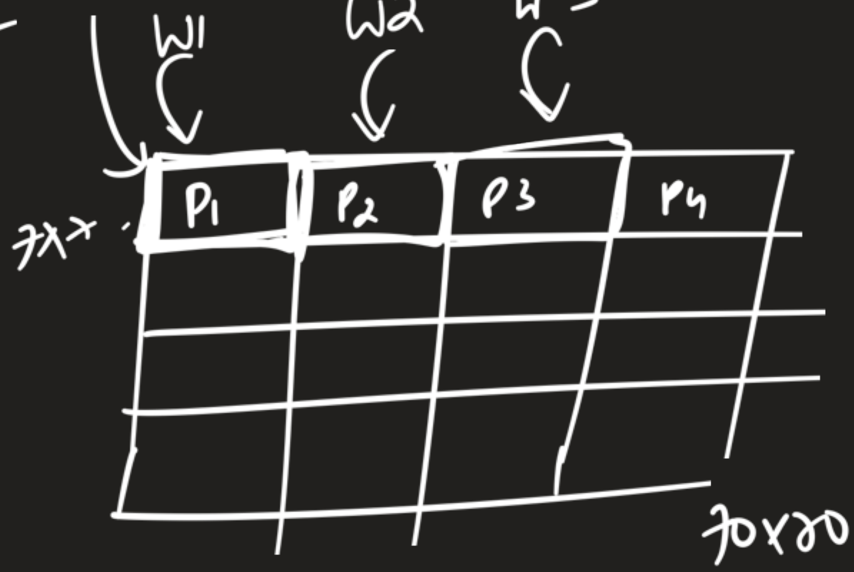
$Q \cdot W_Q \rightarrow Q_p \cdot K^T$

~~7x70~~ 70x70 img

$[\;\;]_{1\times5}$

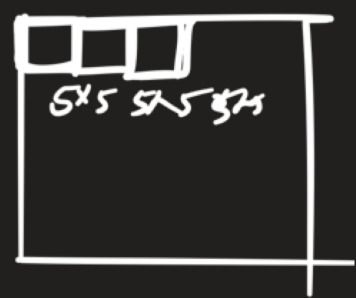
CNN

7x7



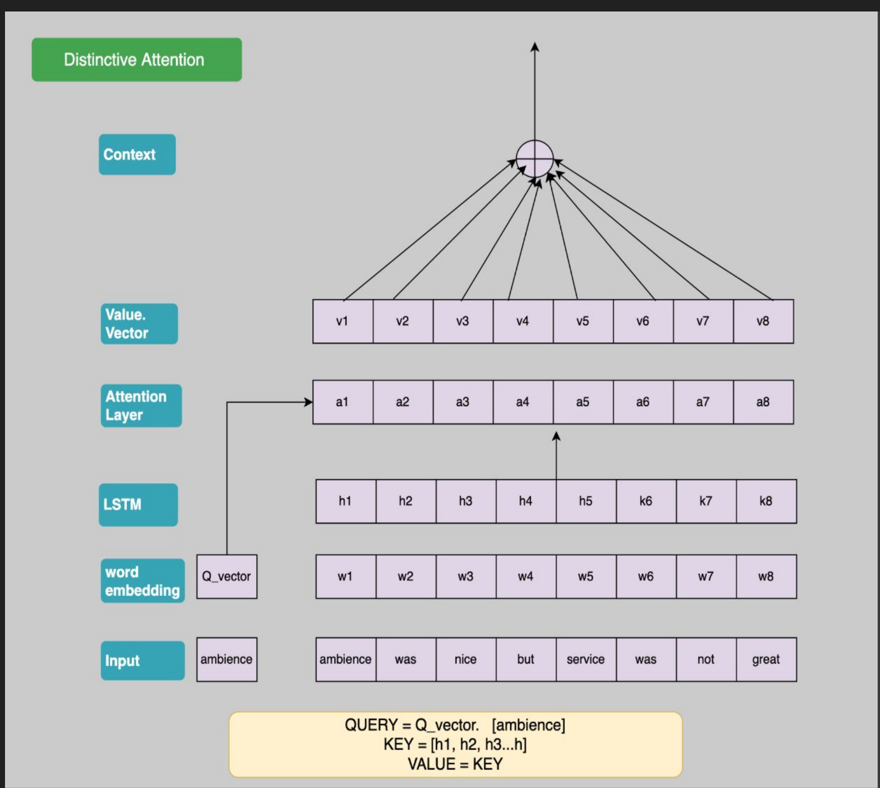
70x70 ⇒

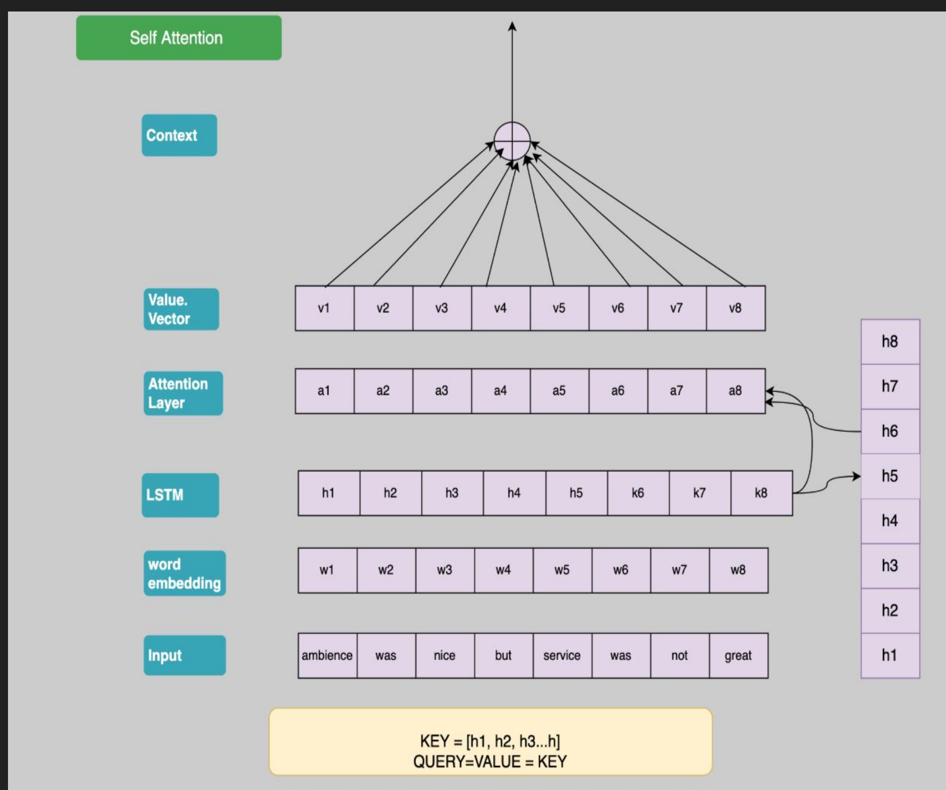
1x7



W1 W2 . . W7

Attention-2


Distinctive Attention
Context
Value. Vector
v1 v2 v3 v4 v5 v6 v7 v8
Attention Layer
a1 a2 a3 a4 a5 a6 a7 a8
LSTM
h1 h2 h3 h4 h5 k6 k7 k8
word embedding
Q_vector
w1 w2 w3 w4 w5 w6 w7 w8
Input
ambience
ambience was nice but service was not great
QUERY = Q_vector. [ambience]
KEY = [h1, h2, h3...h]
VALUE = KEY


Self Attention
Context
Value. Vector
v1 v2 v3 v4 v5 v6 v7 v8
Attention Layer
a1 a2 a3 a4 a5 a6 a7 a8
LSTM
h1 h2 h3 h4 h5 k6 k7 k8
word embedding
w1 w2 w3 w4 w5 w6 w7 w8
Input
ambience was nice but service was not great
h8 h7 h6 h5 h4 h3 h2 h1
KEY = [h1, h2, h3...h]
QUERY=VALUE = KEY

Soft Attention

Attention score is used as weights in the weighted average context vector calculation. This is a differentiable function.

$$z_i = w_i * v_i$$

$$C = \sum_{i=1}^{d_k} (z_i)$$

Hard Attention

Attention score is used as the probability of the i-th location getting selected. We could use a simple argmax to make the selection, but it is not differentiable and so complex techniques are employed.

$$p(s_{t,i} = 1 | s_t, v) = w_{t,i}$$

$$s_t{}^n \approx Multinoulli_L(w_i{}^n)$$

where s_t is a one hot encoder vector which is $= 1$ at position i

Soft

food

P_1 P_6

lunch was good price was high

$P_1 * E_1 + P_2 * E_2 + \ldots$

food

0.9

lunch was good price was high

$[0.9 * lunch_{Emb}]$ content vect

One head

→ one aspect

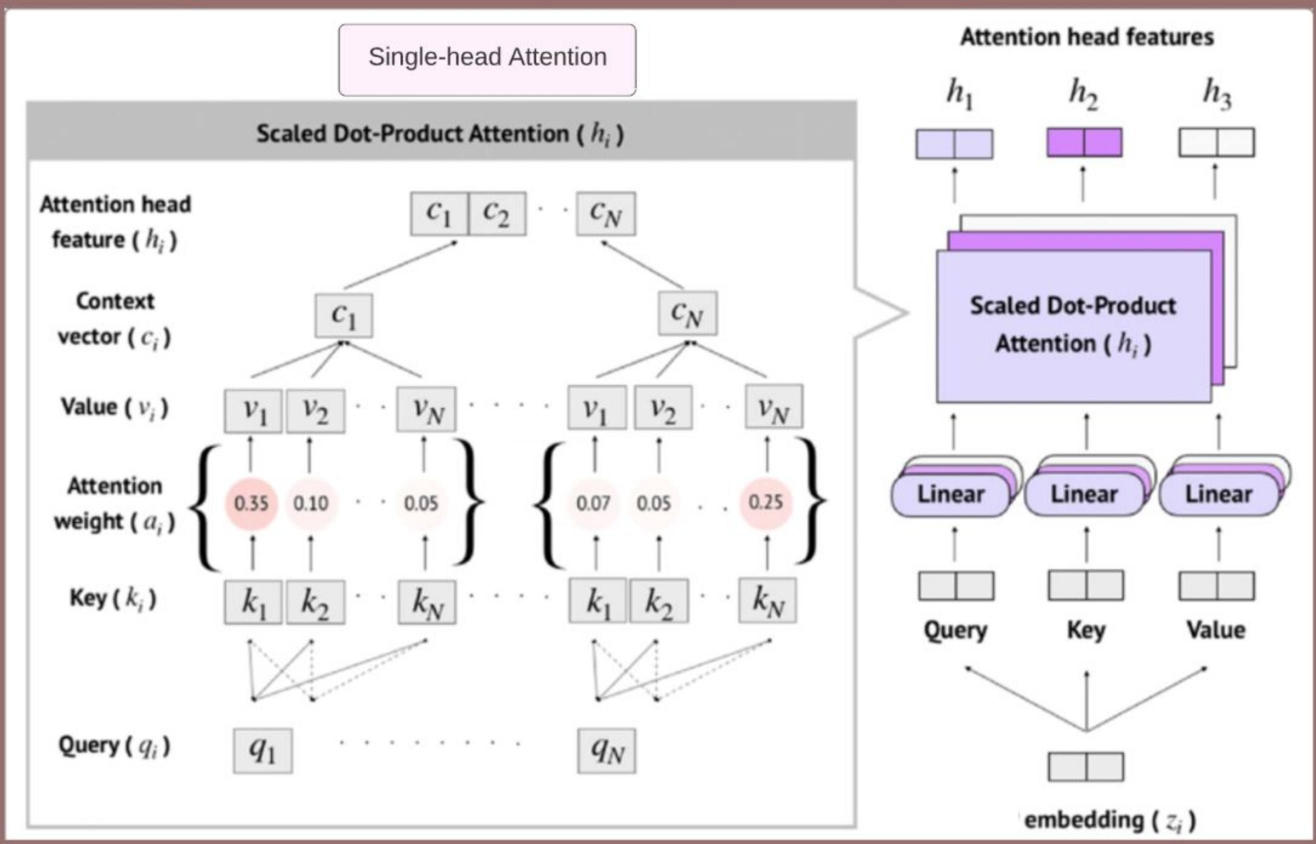
multihead

H1 → grammar

H2 - spatial relation

H3 - semantic

Multi-head attention mechanism with 3 head



3 Linear Layer in right is basically represents KEY, QUERY and VALUE Matrix

~~0.44~~

food — lunch

0.88 good

Multi head attention

1. Step-1 Randomly initialized wq, wk and wv MATRIX.

$Q = K = V$ [Self Attention]

sentence = 11 words

each word = 5 d emb

} Q K V

Q 11x5 . W_Q 5x9 = 11x9

K 11x5 . W_K 5x9 = 11x9

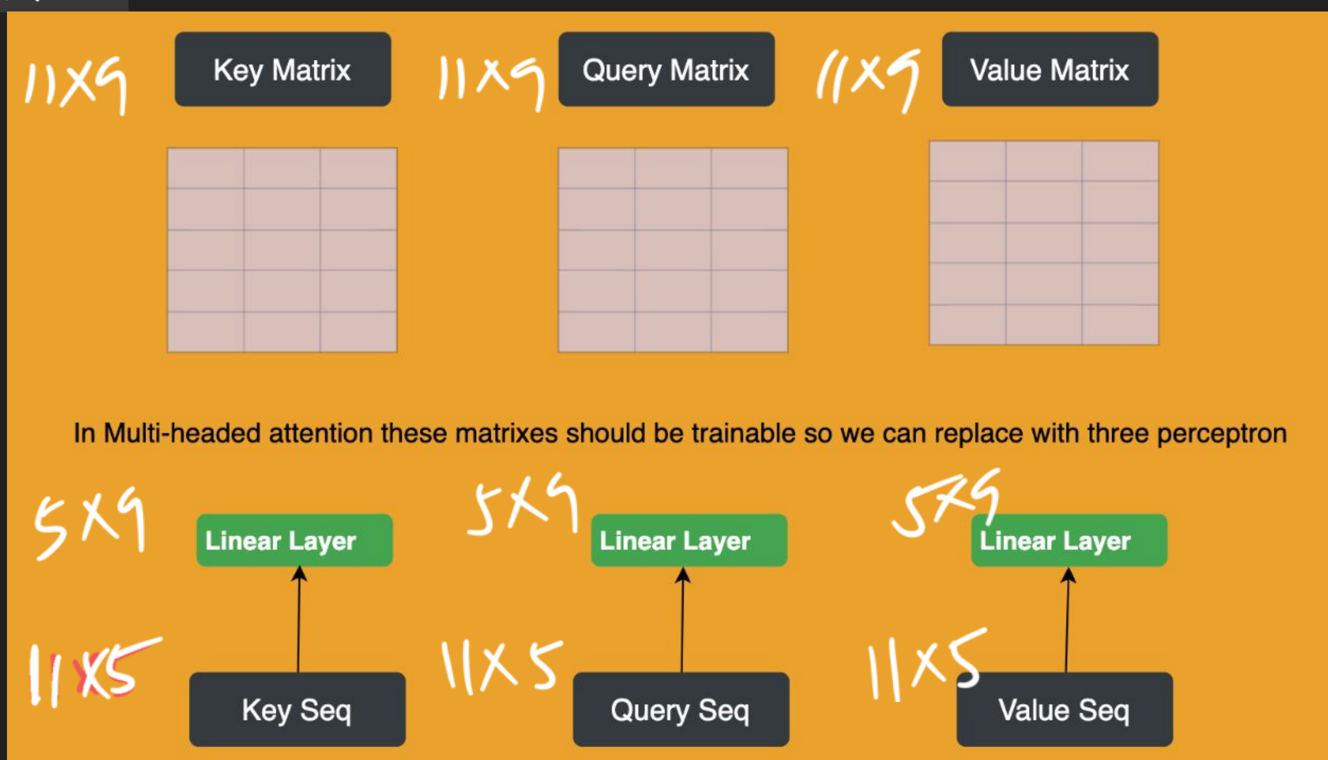
V 11x5 . W_V 5x9 = 11x9

} Q/K/V

learnable trained

2. Define number of Attention Head ← no. of attention heads = 3

3. Compute Query, Value and Key Matrix



$K_{11\times 9}$ $V_{11\times 9}$ } — split into 3 Heads

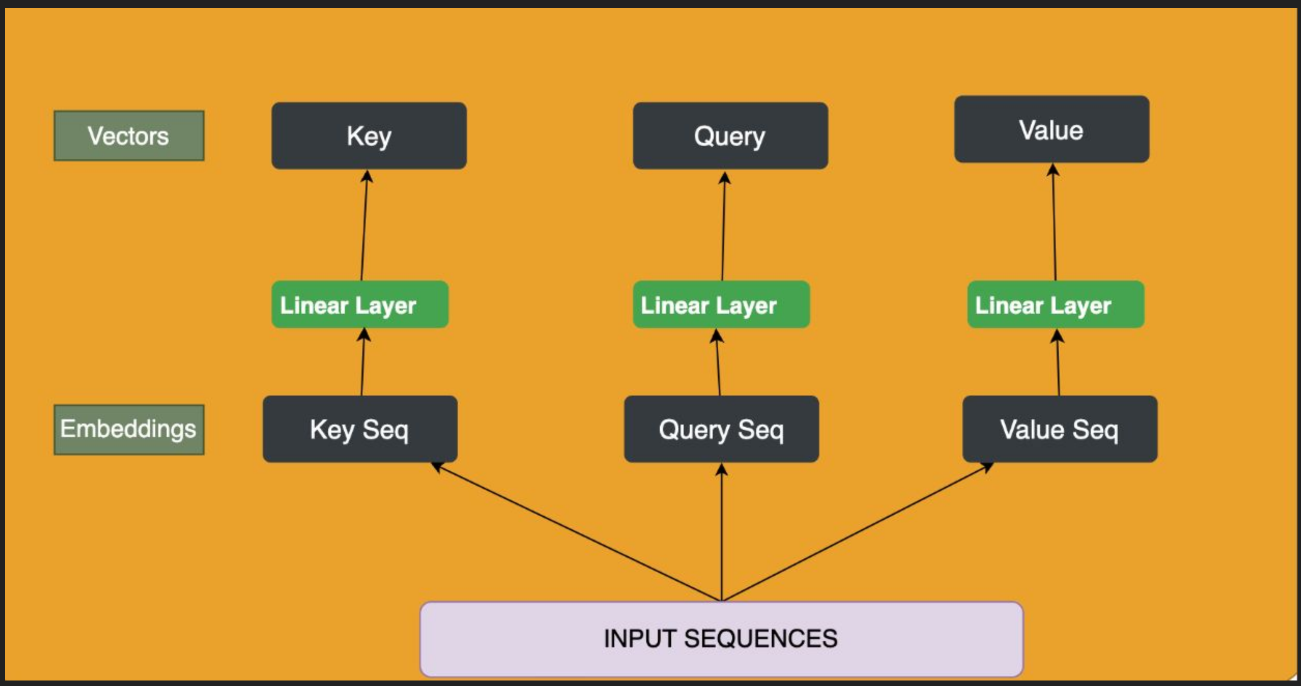
4. Split the Matrix

$Q_{11\times 9}$

H1 11×3

H2 11×3

H3 11×3



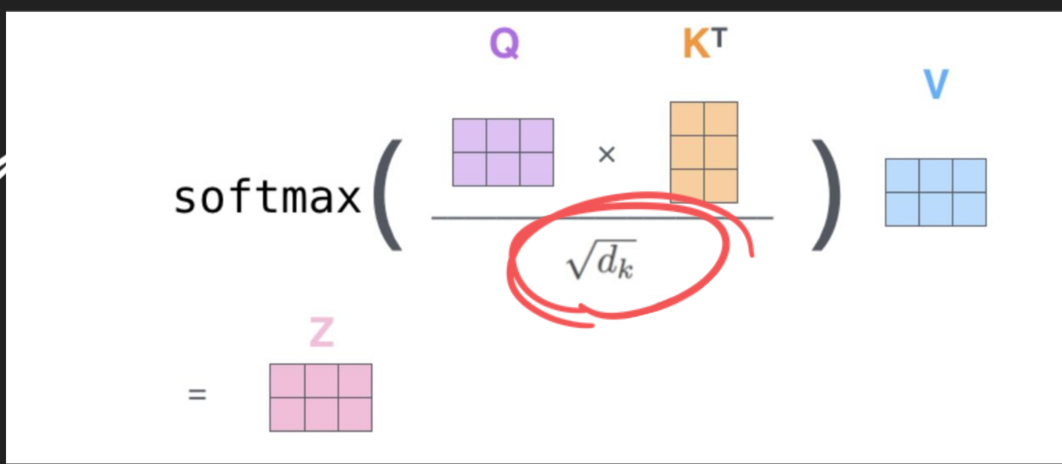
~~11x9~~

$Q \cdot K^T$

$11 \times 3 \quad 3 \times 11$

5. Select an energy function and Compute Attention Scores.

3 such $\begin{cases} Q & 11 \times 3 \\ K & 11 \times 3 \\ V & 11 \times 3 \end{cases}$



how many times — 3 for each head

Context

how many context vectors = 3 [concatenated]

3x11x3

Km[0] → 11x3

11x3

11x3

6. Calculate Context Vector

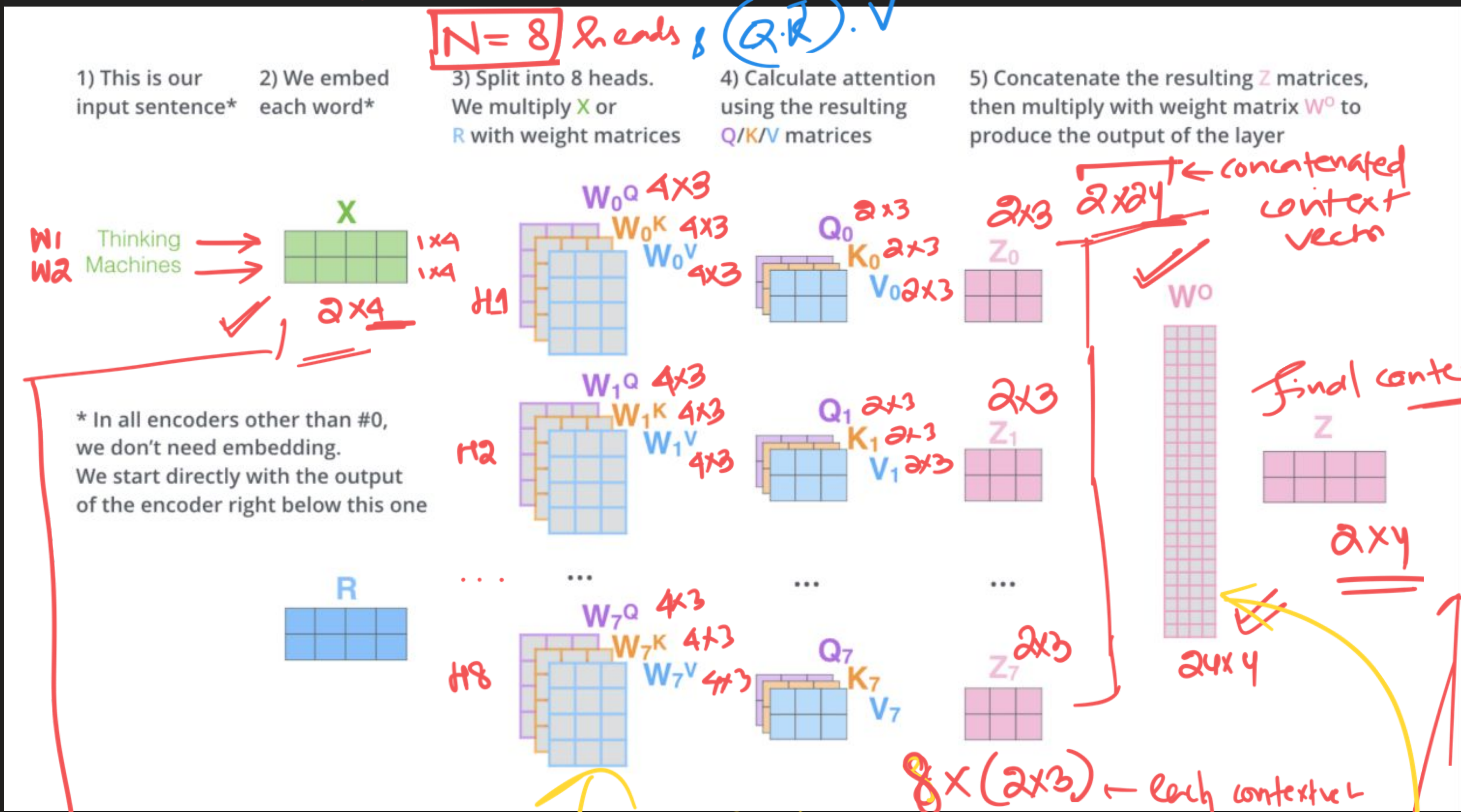
$2\times4 \cdot W_{Q\,4\times24} \rightarrow$ Query matrix $2\times24 \rightarrow 8\times(2\times3)$ }

Heads

$2\times4 \cdot W_{4\times24}$ W^Q W^K W^V 4×24

$\text{softmax}(Q\cdot K^T)\cdot V\,] = 2\times3$

$Q_0\ 2\times3$ $K_{0\,2\times3}$ $V_{0\,2\times3}$



$8\times(2\times3)$ Q/K/V ↑ attn head

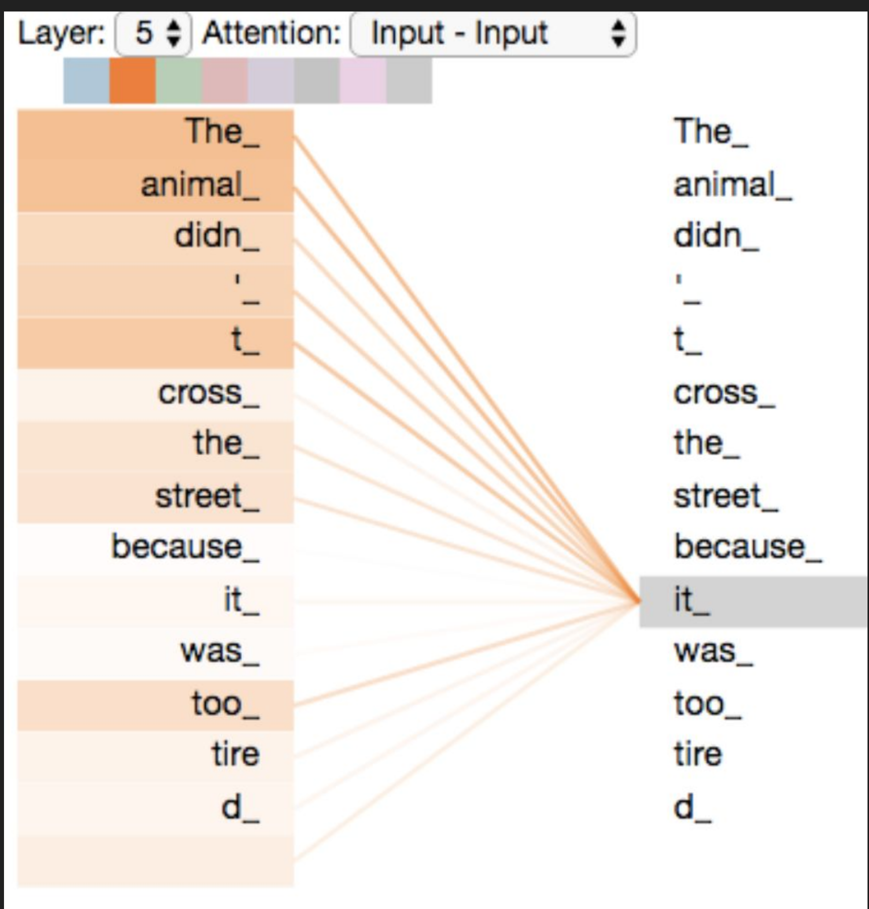
backprop grad

getting updated in train

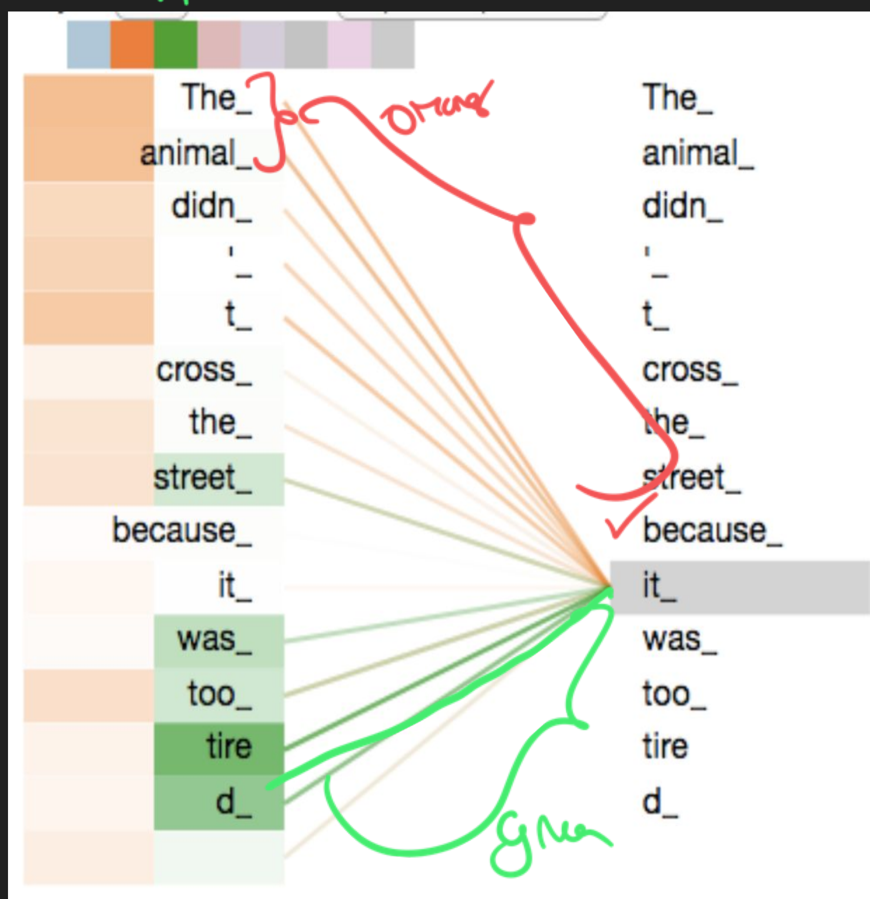
2×24 W_o 24×4 2×4

backprop

single head

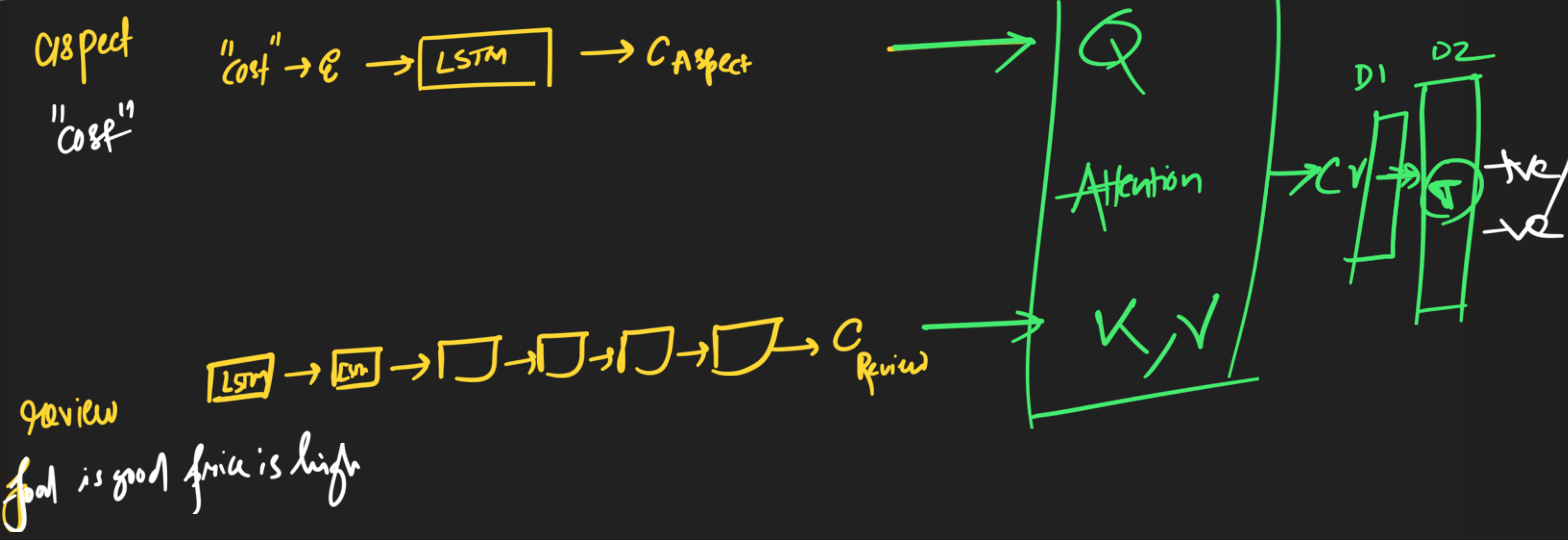


H1
noun



H2 - adj

LSTM + Attention



$N \times S \times V$

no. of sentences

no. of words in a sentence

size of word repn (embedding size)

✓ $10 \times 100 \times 20$

10 sentences

each sentence of 100 word (pad)

Batch - S1 I am good <PAD>

S2 That is so great

Attention mask

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}_{S1}$$

S2

that is so great

that is so great

I am good PAD

I am good PAD

$\overbrace{\text{self.Q}}^{W}$(queries)

queries . WQ

1x5 . 5x9

atmosphere $\longleftrightarrow$ atmosfer

Q $\qquad$ K

W_Q $\qquad$ W_K

S1 I am good

S2 I good <PAD>

$$
\begin{array}{cc}
 & \begin{array}{ccc} W1 & W2 & W3 \end{array} \\
\begin{array}{c} S_1 \\ S_2 \end{array} & \begin{bmatrix} \underset{1\times5}{\boxed{\,\,\,\,\,}} & & \\ & & \end{bmatrix}_{2\times3\times5}
\end{array}
$$

$$
\text{softmax}(Q \cdot K^T) = \begin{array}{c} S1 \\ S2 \end{array} \begin{bmatrix} -11 & -12 & 11 \\ 11 & 12 & \cancel{-11}\ 0 \end{bmatrix} = \begin{bmatrix} 0.8 & 0.2 & 0.1 \\ 0.8 & 0.2 & 0 \end{bmatrix}
$$

$$
\begin{array}{c} S1 \\ S2 \end{array} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}
$$

PAD has no attn wt